

**ENGINEERING
PORTFOLIO
2025-2026**

THE KNIGHTS OF NI

TEAM 5206

DECODE
PRESENTED BY RTX

INTRODUCTION

We are the Knights of Ni, a robotics team associated with the Stanford Online High School (OHS), with **team members around the world**. We were founded in the 2011-12 season and have excelled ever since, even qualifying for the Houston World Championship in 2015. This year, we believe that we have accomplished our goals of **building technical knowledge, empowering our community, and building a unique and innovative robot**.



OPERATING AS A GLOBAL TEAM

As a team associated with an **online school**, our members are geographically dispersed across multiple time zones but concentrated in Northern California. Even within NorCal, some members must drive as much as 1.5 hours to attend in-person meetings. Consequently, we **meet in an onsite-offsite model, which allows members to contribute remotely**.

On Friday afternoons, we meet on Adobe Connect, a teleconferencing platform. During this short meeting, we handle higher-level tasks, such as outlining prototype concepts, distributing tasks, or developing our game strategy. The online format ensures that all members can be involved in important decisions. On Saturdays, we meet for 5-6 hours in-person at a site in the Bay Area. There, on-site members prototype, assemble, and test the robot, while offsite members can call in to work on software and design.



MEMBERS



RACHAEL
Captain



AANYA
Vice Captain



MAXWELL
Software Lead



HAN
Hardware Lead



AYDYN
Outreach Lead



ALEXANDER
Software
EN Lead



WYNN
Software
EN Lead



KATHERINE
Software



ARYAMAN
Software



THOMAS
Hardware

STRUCTURAL SUSTAINABILITY

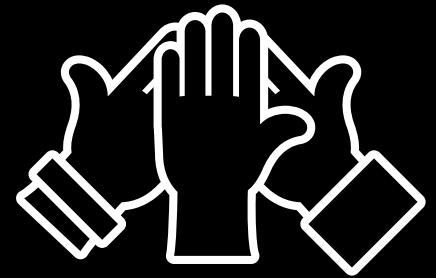
Our organizational structure is arranged in a way that maximizes **stability and sustainability**. The team captain serves for at least 2 years to maintain continuity. The vice-captain plays a key role in transitions, serving a term that is staggered from the captain's. Thus, at least one of the two will have prior leadership experience at any given time, ensuring that hard-earned experience is not lost.

We focus extensively on **developing the skills of new members**, as another means of ensuring sustainability. **Underclassmen/middle-schoolers often hold leadership roles as well**, "owning" a particular mechanism or system, and frequently become sub-team leads. At the start of each season, we dedicate around a month to training, where returning members introduce new members to the basics of FTC and give them a chance to practice their technical skills. This hierarchical system opens doors for new members to research robot or coding systems, be actively involved in prototyping, and build strong technical foundations and soft skills during the season.

Mentors are often parents of current team members who work in relevant Silicon Valley industries. They assist by **providing technical experience and expertise, and sponsorships are often obtained through their companies**.

MISSION

EMPOWERING STUDENTS
THROUGH HANDS-ON
ENGINEERING
EXPERIENCES,
PRIORITIZING LEARNING BY
ENCOURAGING RISK-
TAKING, EMBRACING
INCLUSIVITY, AND
FORMING BONDS BETWEEN
MEMBERS



TEAM VALUES



The main objective of our team is **learning**, so we believe it is important that students do as much of the work as possible. All key design decisions and build processes are **student-led**, while mentors provide advice if needed. The main objective of our team is learning, so we believe it is important that students do as much of the work as possible. Our team hopes to learn and educate through hands-on experience in engineering. This process-oriented approach heavily influences our design decisions; we are more likely to select a unique and interesting mechanism as we prioritize the valuable learning lessons that come with taking these **risks**.

We also pride ourselves as a team open to anyone—it is our policy to not turn anyone away due to lack of experience. Instead, we put a heavier emphasis on **skill-building** through CAD and coding workshops to raise our members to a sufficient level of expertise. Members with varying degrees of experience are encouraged to collaborate with one another, fostering an environment where **inclusivity** and learning work hand in hand.



GOALS



MEASURING SUCCESS

IMPACT OVER OUTCOME

Whether it be inspiring FLL teams or learning a valuable lesson about a new mechanism, we are more appreciative over the journey than an objective success.

STRENGTHENING COMMUNITY

Both within the team and outside, we prioritize robotics as a community, rather than a competition.

INCLUSIVITY

We firmly believe that if anyone has an interest in robotics, it is our mission to cultivate that interest by providing resources.

TEAM ORGANIZATION PLAN

== Team Bonding ==

Virtual movie nights, game nights, and escape rooms

== Onboarding ==

Create a Notion page to introduce new members to the team

== Individual Goals ==

Have members set clear goals for themselves, and help one another achieve them

== Local Community Service ==

Reach out to local libraries and schools for demonstrations, workshops, etc.

== Documentation ==

Consistently write meeting minutes, scheduled deadlines, and task sheets to keep track of progress

OUTREACH



As an online team spread across the world, our team sought to engage in virtual and **sustainable** solutions to gaps in our respective **communities** around the globe. We planned outreach events at the intersection of FIRST values and robotics education. We also epitomized Gracious Professionalism on and off the field by **mentoring** other FIRST robotics teams to make robotics more **accessible**. Overall, our outreach efforts **impacted dozens of students**.

Each experience has only strengthened our values, and **their results are tangible in our team culture**. For example, mentoring FLL teams has helped veteran members learn how to mentor new members; reporting on FIRST competitions in our school magazine has helped us introduce robotics even to people who aren't already in STEM.

CAD TUTORIALS



We created a **publicly accessible** website with a comprehensive set of Fusion360 tutorials. This website was shared by the NorCal FTC on their website, the NorCal FTC Teams Helping Teams Resource Page. Our tutorials draw upon our personal experiences working in FTC. They aim to combat the deficits in mentorship through **step-by-step guides and challenge problems** that both rookie and veteran teams can take advantage of.

SUPPORT THROUGH DONATION

We donated four boxes of FLL parts to support an FLL team in Austin, Texas. This donation allowed for **an entire FLL team to have individual robots for each member**, allowing students to be able to test ideas individually and then collaborate as a group as well. Altogether, this allowed the team to be able to work both on teamwork and individual skills.



TUTORING

Members individually tutor **underserved students** in K-5 STEM & Humanities. We tutored and designed Python curriculum for English Language Learners.



OUTREACH

STEM COURSE SERIES

Whether it be linear discriminant analysis or the Gram-Schmidt process, our STEM Course Series shares knowledge on a broad variety of STEM topics, like linear algebra. With many members of our team taking **university-level courses**, we aim to decrease the barrier for high school students to access such **resources**. Rather than explaining topics in dense technical terms, we brought in real-world motivations to make engagement accessible.

IRIS EXAMPLE

Class	μ_{PL}	μ_{PW}	N
Setosa	1.5	0.2	50
Versicolor	4.3	1.3	50
Virginica	5.5	2.0	50

Given the class means, compute the overall mean:

$$\mu = \left[\frac{1}{150} \sum_{i=1}^{150} x_{i,PL} + \frac{1}{150} \sum_{i=1}^{150} x_{i,PW} \right] = [3.76, 1.16]$$

Compute the within-class and between-class scatter matrices:

$$S_W = \begin{bmatrix} 0.22 & 0.08 \\ 0.08 & 0.06 \end{bmatrix} \quad S_B = \begin{bmatrix} 2.22 & 1.12 \\ 1.12 & 0.62 \end{bmatrix}$$

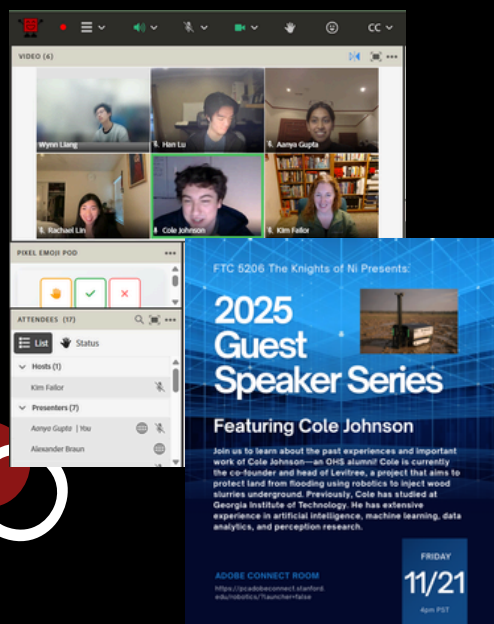
Compute the eigenvectors and corresponding eigenvalues for the scatter matrices:

$$S_W^{-1} S_B w = \lambda w$$
$$S_W^{-1} = \begin{bmatrix} 4.35 & -5.8 \\ -5.8 & 15.9 \end{bmatrix}, \quad S_W^{-1} S_B = \begin{bmatrix} 5.2 & 3.1 \\ 3.1 & 2.6 \end{bmatrix}$$
$$\lambda_1 = 7.1, \quad \lambda_2 = 0.7, \quad w_1 = \begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix}, \quad w_2 = \begin{bmatrix} -0.6 \\ 0.8 \end{bmatrix}$$

iris setosa iris versicolor iris virginica

petal sepal petal sepal petal sepal

SPEAKER SERIES



We invited multiple speakers to our school, ranging from individuals in academia to industry. We started by bringing our past vice captain, Thomas Lu, a current physics and computer science student at Berkeley. Thomas **shared the lessons he learned in participating through FTC** and its impact, not only in college, but in his identity.

In addition, we invited Cole Johnson, an alumnus of our school. He is both a researcher and entrepreneur from Georgia Tech and Caltech. From Cole, the team learned a lot about the **applications of robotics** in other fields of study.

These speaker events allow us to ask questions and potentially even **form new mentors**, as well as **engage with other STEM communities**.

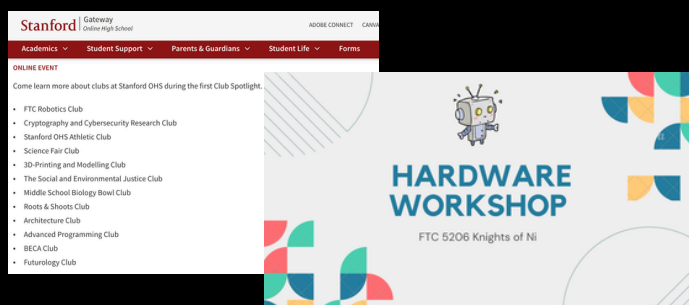
PEER TUTORS AND TA'S

At Stanford Online High School, **half of our team members are peer tutors and/or teaching assistants**, including Anya, Rachael, Aryaman, Alexander, and Maxwell. Peer tutors and teaching assistants are selected at the completion of the previous school year, having exemplified the OHS values of academic achievement and **service to the school community**. They commit at least two hours per week of appointments, as well as hosting weekly office hours and exam review sessions.

OUTREACH

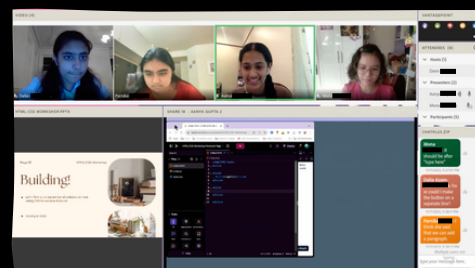


SCHOOL WORKSHOPS



Each year, as the anticipation for the season builds, we **host school-wide hardware and software workshops open to all students**. These workshops attract dozens of students. Not only are we able to promote membership towards joining our team, but FIRST as a whole too.

Some of our members are ambassadors for our school's **Girls Can Code Club**, to spread our joy for STEM without communities. This year, we held 2 workshops for students with any coding background. Students learned from the very basics on how to **build websites using HTML/CSS**.



We collaborated with the Madam Math Circle, including students in grades 3-6. We discussed common math terms and theorems applicable in robotics. Further, we guided students through the application of math for this year's FLL mini Mission, Light Show, hoping to make them the next generation of FTC robotics!

MENTORING



With international roots, we place a strong emphasis on **global outreach**. Therefore, we took up the opportunity to mentor an FLL Team at High Gate International Academy in Kigali, Rwanda. Each week, we meet with the team to answer any questions they may have, provide guidance on their innovation project, and help them find the same spark we found as young FLL Roboticists. We also aided in the creation of a second team in Austin, Texas. These teams were initially just groups of students interested in robotics, but we helped them solidify and explore their interests by **transitioning them into FIRST.- 49 Hours**

CLUBS

Many team members lead or are ambassadors of school clubs: the Science Bowl Club, Advanced Programming Club, Logic Olympiad Club, Science Research Club, school magazine, 3D Printing and Modeling Club, and Mathematical Modeling Club. These clubs **promote STEM topics through workshops/speaker series, competitions, writing news articles about FIRST, and more.**

DESIGN PRINCIPLES

Due to our constraints as an online high school, our design team prioritizes three values:

EASE OF ASSEMBLY AND MAINTENANCE

To optimize our time in-person, we aim to minimize maintenance: from choosing a belt-driven viper slide over a string-driven linear slide, to using principles of physics to balance any pressure on servos.

BEING ROBUST

As a team that strives for growth, we ensure our designs are robust—whether that be making efficient uses of space or coupling functions.

RELIABLE

Since drivers have limited time in-person, parts are intuitive to operate. Thus, we can reduce human error. Moreover, by coupling the function of the arm as both intake and outtake, our design's pass-through is incredibly reliable.

Computer-aided design (CAD) is extremely important to us and our entire robot is constructed virtually in the CAD software **Fusion 360 before it is built**. This allows **off-site members to participate in building** and helps us check for interference and other potential issues before the more costly build process begins.

From **7th to 12th graders**, our team features a large range of ages. As a result, we place an emphasis on **mentorship**. To onboard hardware members, we introduce them to the following process:

Types of design consideration, such as intake or extension

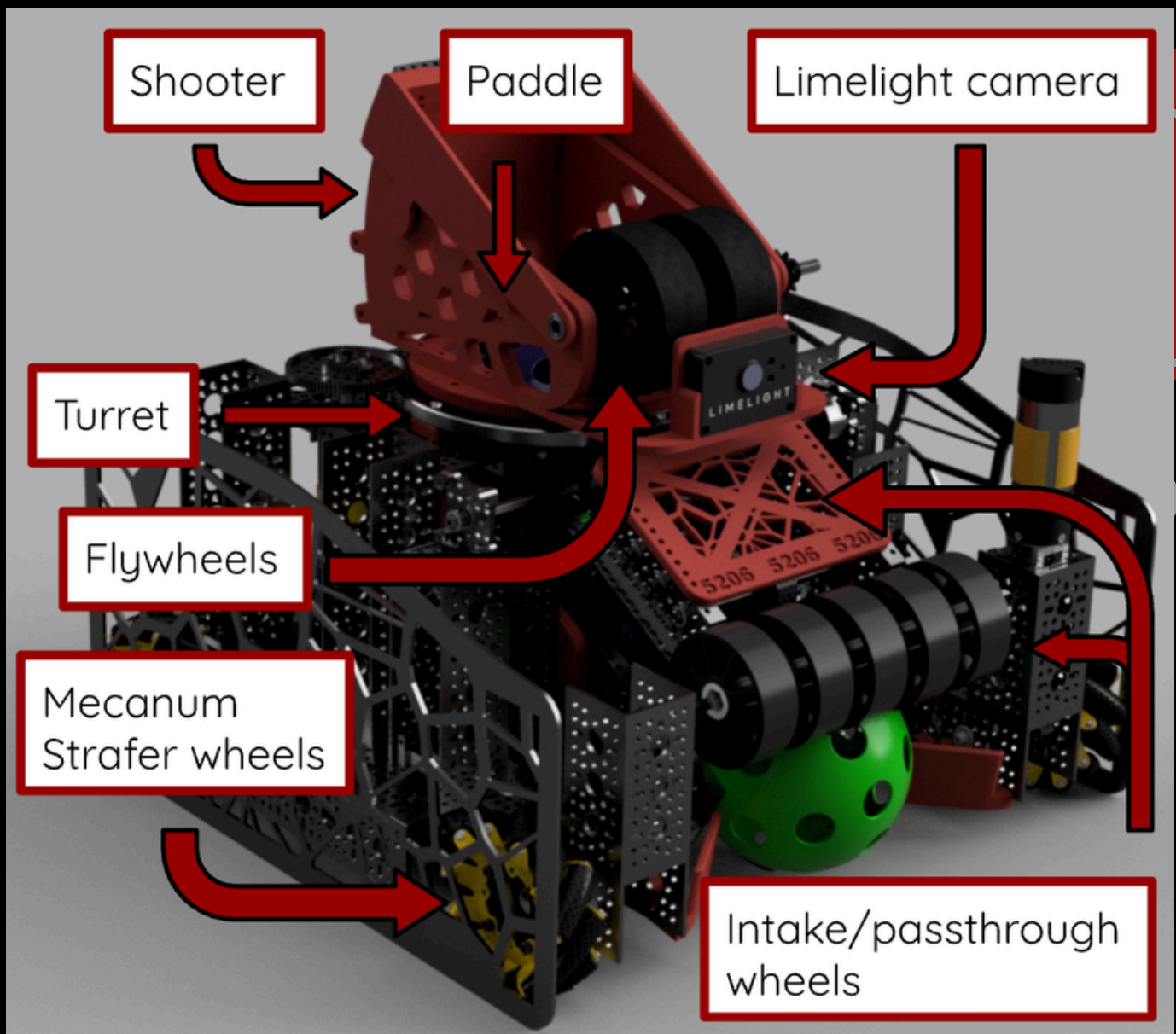
CAD projects, demos, and tutorials

Introduction to 3D printing, using our Prusa MINI+

Training for assembly, maintenance, and iteration

This season, we wanted our hardware components to **minimize cycle time**. This included robust scoring from any orientation on the field. We approached this in two ways: 1) **having a fast passthrough** from intake to shooting, and 2) having a **shooting mechanism that can rotate via a turret**. We also collaborate by having **team-wide discussions** about design considerations—thus building a robot that is not only reliable, but also **compatible** with software and overall strategy.

ROBOT OVERVIEW

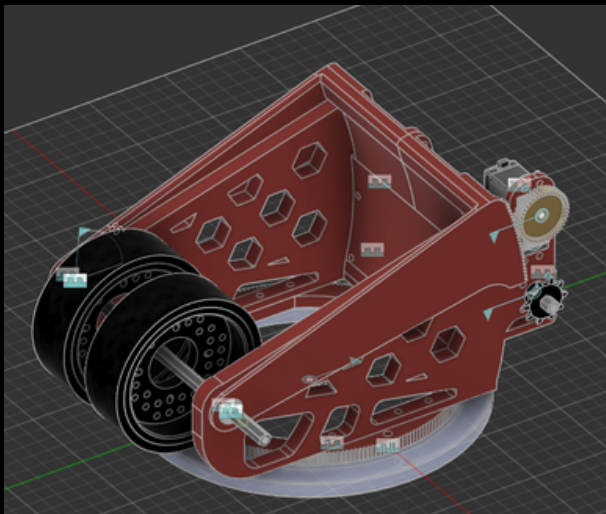
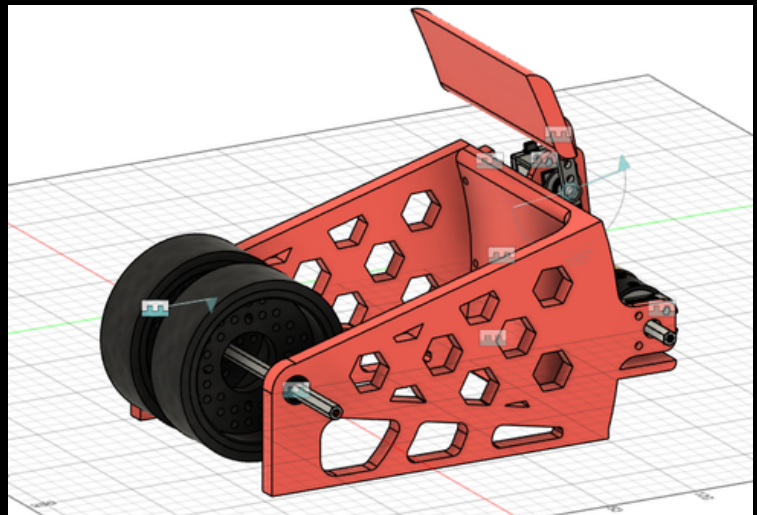


ITERATIVE PROCESS

Design is an iterative process. Throughout the season, our team **considered a variety of prototypes** and focused on improving on our initial design.

CASE STUDY: SHOOTER

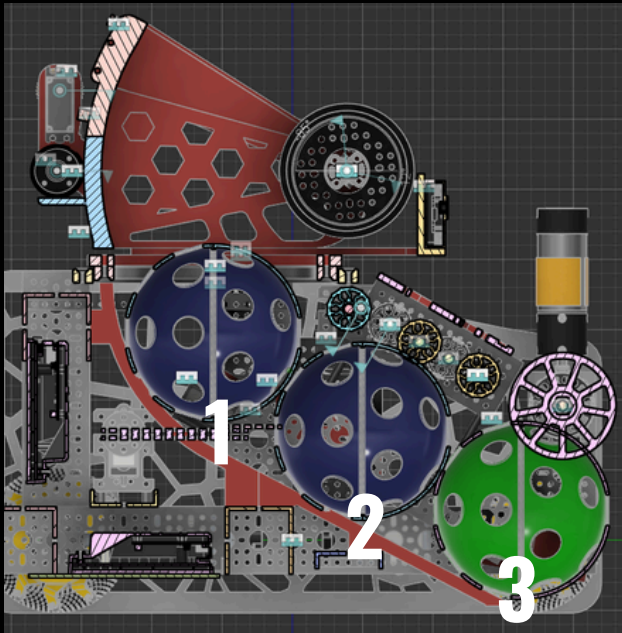
For example, consider the shooter. Initially, we designed **a flap** such that the balls could shoot at varying angles. However, this design had **consistency issues**. The flap was powered by a servo—which couldn't counteract the high velocity of the ball, causing it to break. The angle would also slightly vary upon impact with the ball (despite the servo being powered), leading to inconsistent shooting. Moreover, even the **highest angle couldn't shoot far enough**.



As we iterated on the design, we replaced the initial hood with a new one that had **more degrees of freedom**. We could now shoot from anywhere—both far and near. To improve **shooting consistency**, we first lined the hood with plastic to increase traction and friction, then added weights on the rotating wheels to increase momentum. Though this design improved consistency issues, it didn't resolve them entirely. At this point, we realized that a **servo-powered flap wasn't compatible**. We instead took advantage of varying the flywheels' motor power through software for robustness.

COMPONENTS

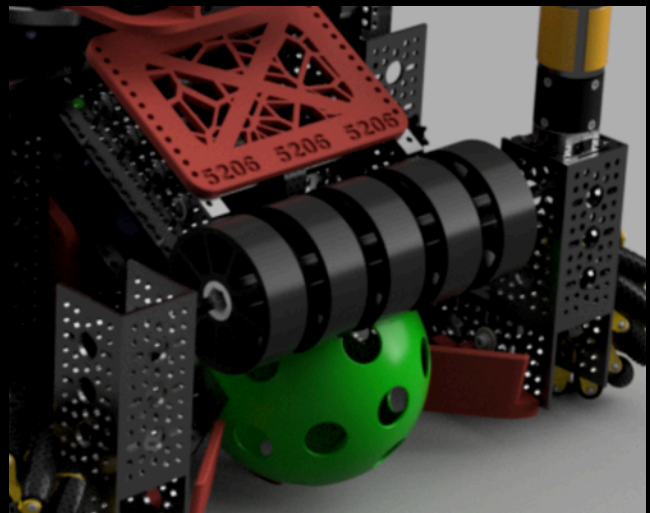
PASSTHROUGH



However, we learned that greater speed meant **less control** over the artifacts—that is, ensuring they didn't just immediately get pushed into the shooter (and randomly fire) or get stuck in the second set of wheels. Therefore, we **iterated on our design** by use a sequence of “paddles” for greater precision.

Specifically, we were only able to shoot artifacts in positions 1 and 2 (shown above), ending up with the initial artifact in position 3 stuck at position 2. With space and power constraints, the set of passthrough wheels were unable to transfer the last ball to the paddle. Now, the second paddle brings the artifact at position 2 to position 1, so we can shoot all 3 artifacts at once.

Intaking the artifacts involves sets of intake wheels which **roll the artifacts into the robot**. Once within the robots frame, they are **pushed by a paddle into the shooter**. We chose this design to minimize the time spent transferring the ball from the field to the shooter. In the same vein, we chose not to implement an indexing system because we believe the strategic tradeoff from MOTIFS is simply not worth it; **decreasing cycle time** in TeleOp was our priority.

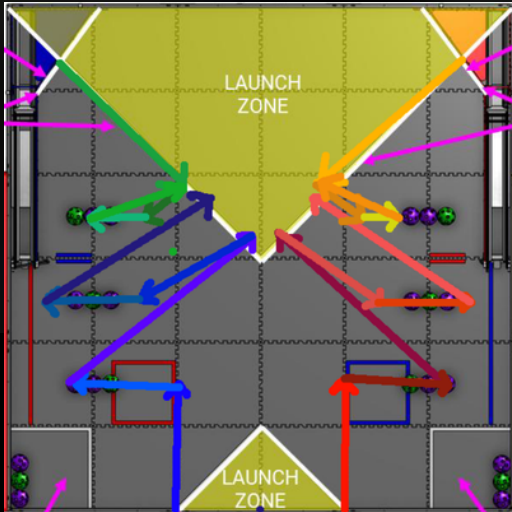


SHOOTER

For our robot, our team used a flywheel—a spinning wheel—to launch game elements **quickly and consistently**. A fast-spinning mass stores rotational energy (angular momentum), which transfers to the projectile when it contacts the wheel, so we needed to have a fast flywheel to make our projectiles go high and far.

GAME STRATEGY

AUTO



In all of our autonomous paths, we score artifacts in the **goal**. We aim to **prioritize iteration** in autonomous, ensuring the maximum number of artifacts scored.

In top blue/red we score 3 samples in the goal, then cycle to score another 3 samples. In bottom blue/red we score 3 samples in the goal, then pick up 3 more.

To improve reliability during this period, we integrated the Limelight camera to set accurate motor power, as well as odometry for drive (described on next page).

TELEOP

In TeleOp, our intake collects artifacts and delivers them to the turret for scoring. The turret uses **Limelight and AprilTag detection to aim automatically**, ensuring smooth and accurate shooting while the driver focuses on positioning the robot. As the distance from the AprilTag increases, the power with which we shoot increases as well.



Below is an image of our gamepad mappings. We have coded each gamepad to have the **specific functionality** needed for the drivers. **Drivers work together**, controlling different aspects of the hardware. We recognized that the robot had “blind spots” when it was too close to the goal—and thus couldn’t see the AprilTag. As a **failsafe**, we also allow the drivers to manually set appropriate velocities for near and far shooting.

Game Control: Primary Driver



Game Control: Secondary Driver



DRIVE

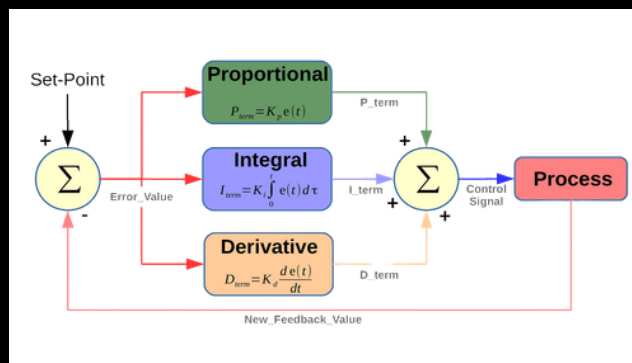
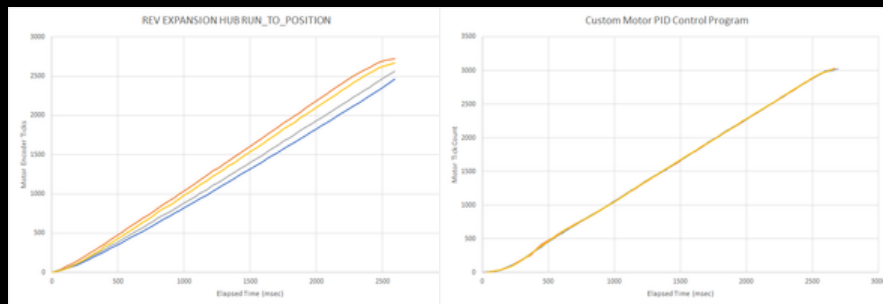
PID

To move our robot consistently and correctly along our selected path, we use PID (Proportional Integral Derivative) control. This aligns our robot's current direction (**kP**), speed (**kD**), and position (**kI**) with our desired states.

- **kP** -> **current error** (where are we currently, where do we want to be?)
- **kI** -> **accumulated error** (what been the outcome of past correction)
- **kD** -> **predicting error** (anticipating future conditions based on current rates of change)

Why do we need PID?

- We found that the robot would not go in a straight line, even though we set identical speed and position targets for all four motors.
- Testing revealed inconsistencies in the rate of all the motors, where the REV internal algorithm does not guarantee that all motors rotate synchronously, leading to potential deviations from a straight line.
- PID helps us ensure that all motors move together and addresses inconsistencies, guaranteeing precise movement.
- Before, our current and desired states looked like the graph on the left.



ODOMETRY



This year, our robot uses odometry to track its location on the map at all times. This lets us **estimate the position of the robot** by recording the odometry wheels travel along the x and y axes.

The use of odometry helps during the autonomous period as the robot is able to **correct itself** to reach the target location, and from there **more accurately score**.

SCORING CONTROL

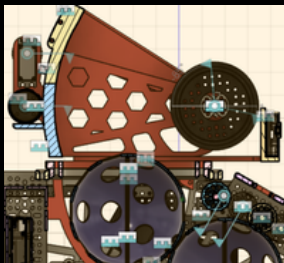
AUTO - AIMING

Our hardware design allowed us to utilize the turret through software to track AprilTags, so the driver can more efficiently work on other tasks. The **Limelight 3A** camera detects nearby tags, and, combined with odometry, provides **accurate pose estimations**. Both the camera and odometry were new systems we chose to challenge ourselves with this year.

We use a custom-made sigmoid function to smoothly adjust the turret velocity for aiming, rotating the turret until the target is at the center of the camera's field of view, **ensuring the turret works reliably and consistently hits its targets**. When the target is too far from the center, we rotate the bot instead.



FLYWHEEL



Distance (cm)	Power
88	0.8
36	0.6
48	0.65
60	0.7
72	0.75
84	0.8
96	0.85
108	0.9
120	0.95

To vary the **velocity** of the flywheel without placing strain on drivers, we again use distance estimations from the **Limelight 3A** camera. First, we **collected data** on the ideal velocity that corresponds to a distance. Nevertheless, **the real velocities didn't match the ideal velocities**—especially when artifacts were quickly and successively shot. The flywheel would initially slow down after each artifact, making shooting inconsistent.

After tuning PID for our drive movements, we realized that it was a **fundamental principle** that could also account for errors in flywheel velocity. Consequently, we implemented a PIDF controller to recover the target RPM immediately. If the distance changes drastically—thus forcing the motor behavior to also change drastically—we account for any potential errors with **PIDF**. PIDF adds a Feedforward ("F") term. The F term predicts the needed output based on the target value, helping the motor reach the target faster and more accurately.

